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Generative Artificial Intelligence for Next-Generation Wireless Networks

THE UNIVERSITY OF HONG KONG 香港大學 • FACULTY OF ENGINEERING 港大工程學院

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NiCE | Network Intelligence and Computing Ecosystem Lab

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- IEEE Daniel E. Nobel Fellowship Award
- IEEE ComSoc Best YP Award
- Singapore Data Science Consortium Research Fellowship
- Editor, IEEE COMST, TCOM, TVT

Homepage

RESEARCH

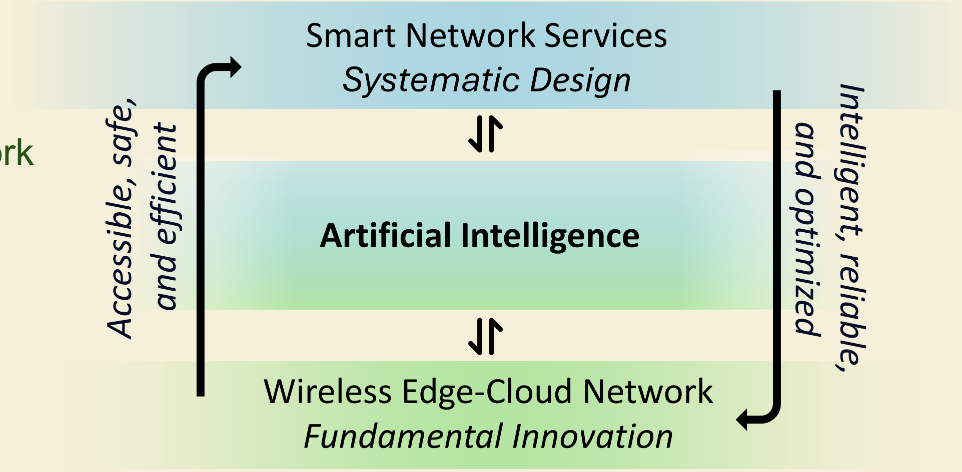
AI for Network:

- Generative AI and network optimization
- Reinforcement learning
- Human-centric AI for user-centric network

Network for AI:

- Distributed training and inference
- Next-generation wireless transmission
- Edge Intelligence

HIGHLIGHTS



- **Generative Diffusion Model-aided Network Optimization:** codebook generation, signal spectrum generation, etc.
- **Large Language Model-aided Network Optimization:** retrieval-augmented generation, RL-aided decision making, etc.

PUBLICATIONS

- H. Du, et al, "AI-Generated Incentive Mechanism and Full-Duplex Semantic Communications for Information Sharing," IEEE Journal on Selected Areas in Communications, 2024. **IEEE JSAC Popular Documents**
- H. Du, et al, "Mixture of Experts for Intelligent Networks: A Large Language Model-enabled Approach" in Proc. IWCMC, Cyprus, May 2024. **Best Paper Award.**

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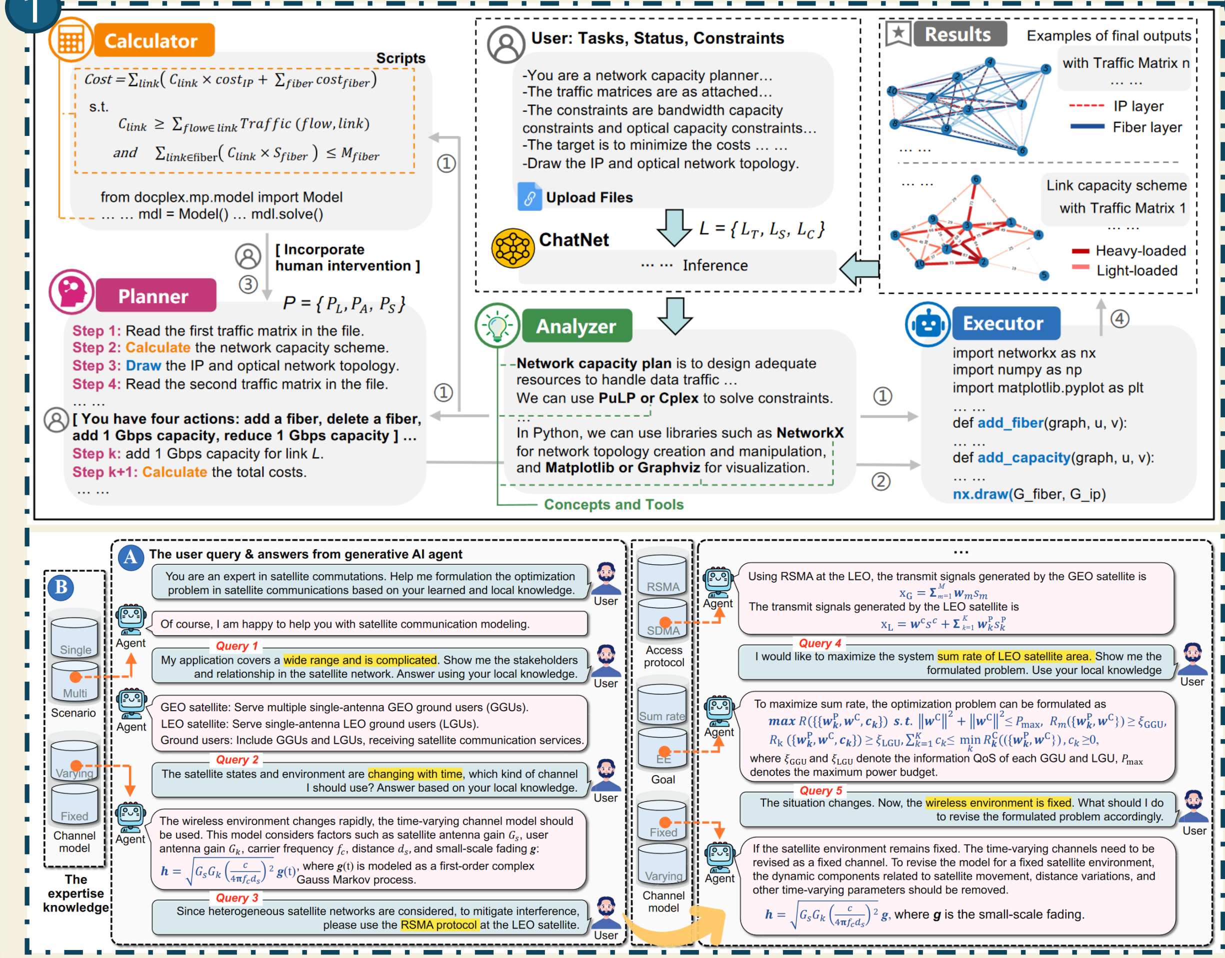
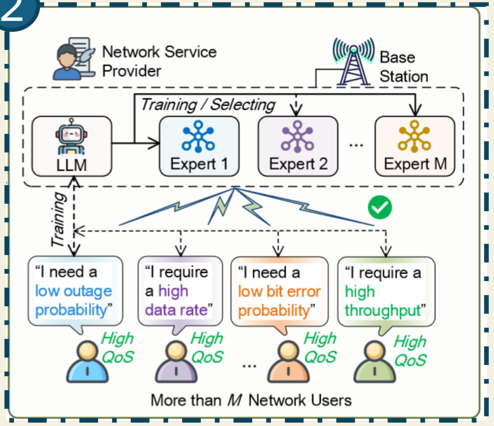
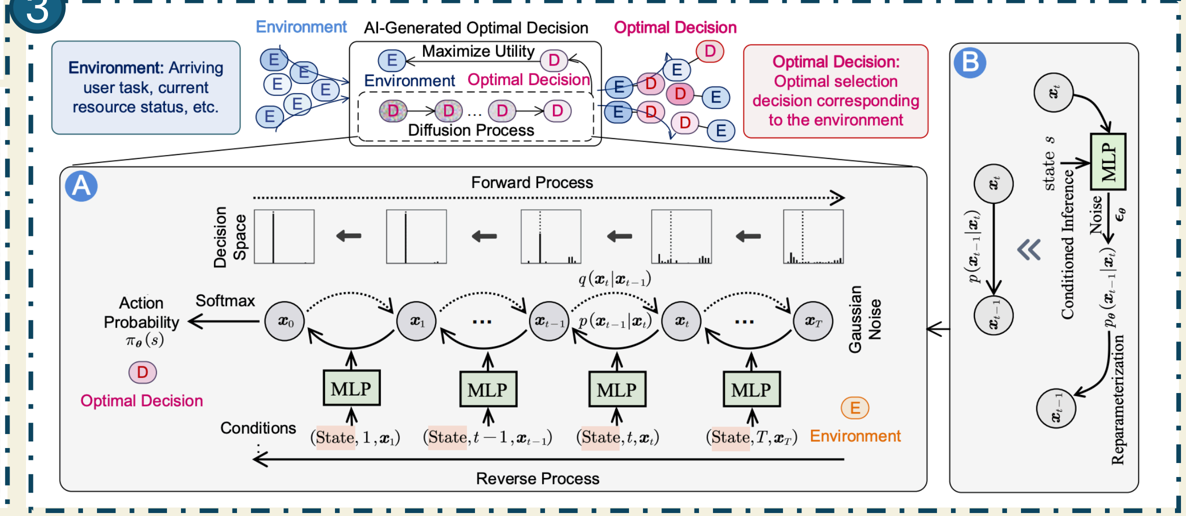
GenAI for Network

We use GenAI techniques, e.g., LLMs and Diffusion Models, to empower network intelligence and automation, enabling a new paradigm of AI-driven network optimization and decision-making.





1. LLMs as Agents for Network Optimization and Management
2. Network Hybrid Expert System for Multi-Task Optimization
3. Diffusion Models for Network Decision Generation



H. Du, et al. "Diffusion-based reinforcement learning for edge-enabled AI-generated content services." IEEE Transactions on Mobile Computing, 2024.
H. Du, et al. "Enhancing Deep Reinforcement Learning: A Tutorial on Generative Diffusion Models in Network Optimization." IEEE COMST, 2024.

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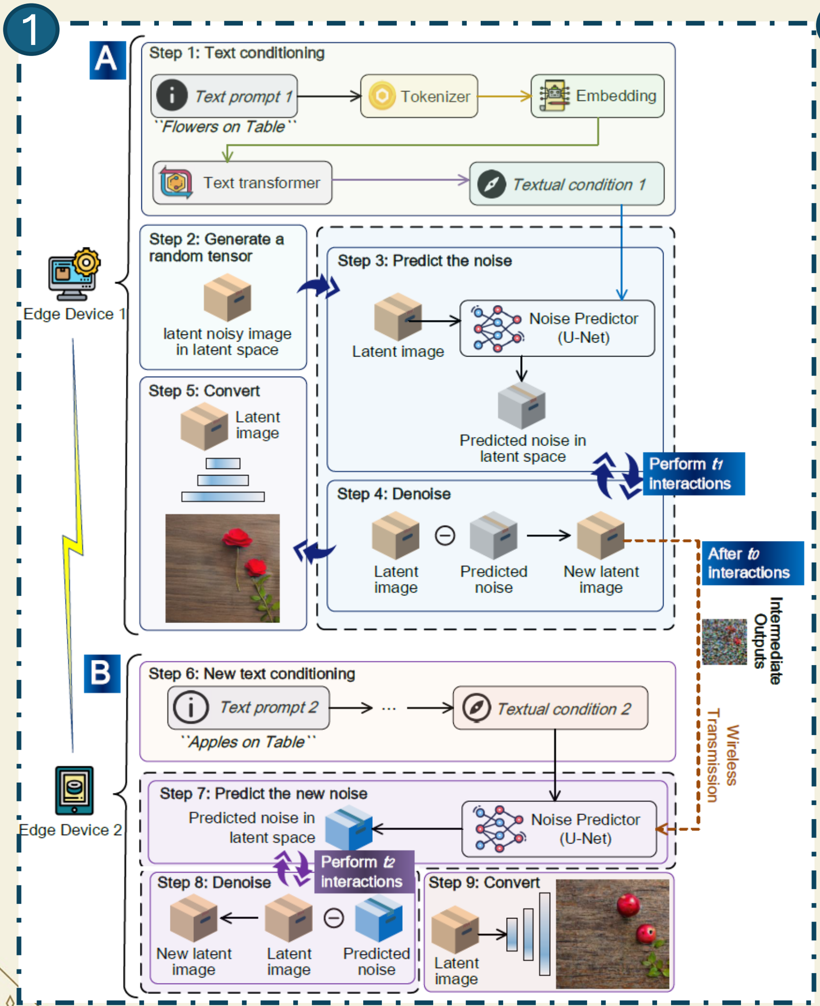
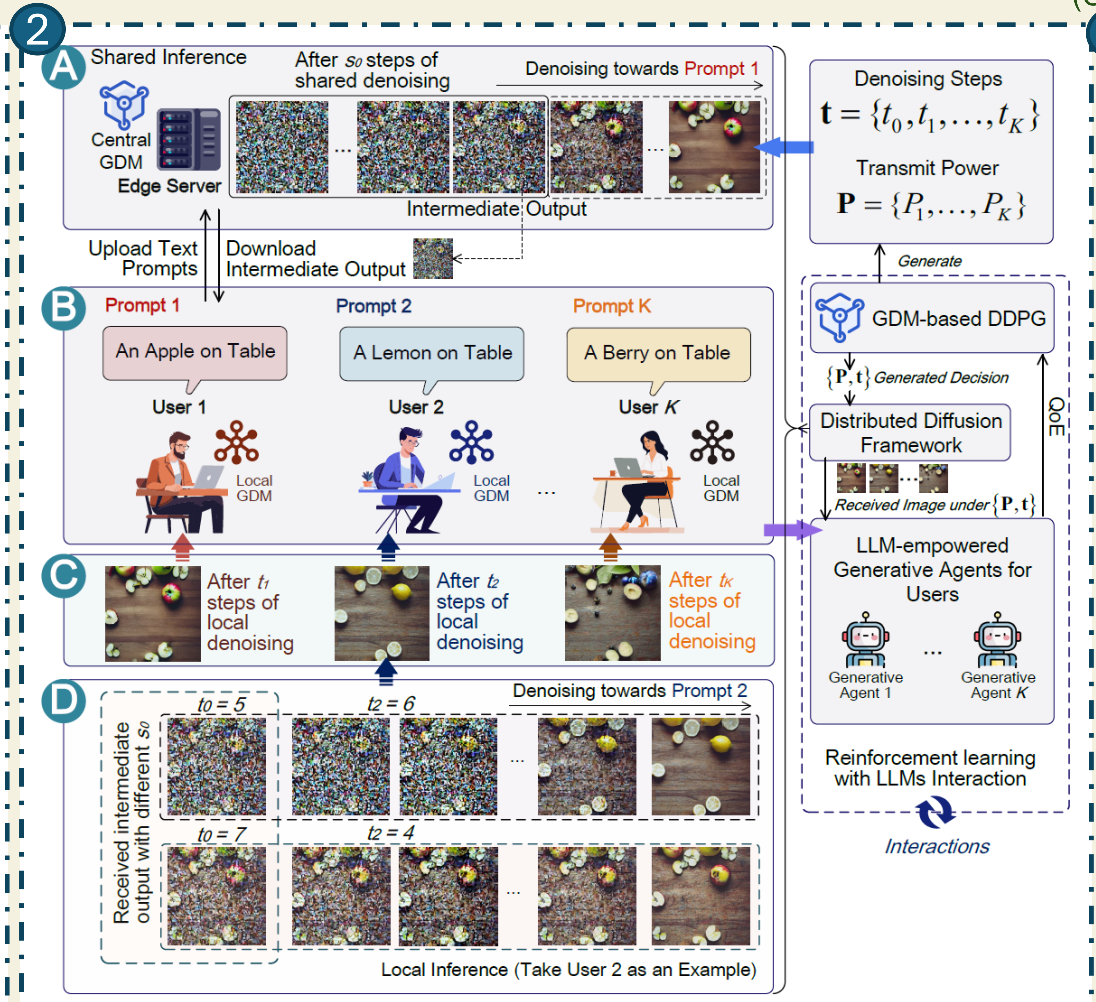
Network for GenAI

We utilize 6G networks to enable flexible delivery of personalized GenAI services, while achieving cross-GPU, cross-device, and cross-data center model training and inference.

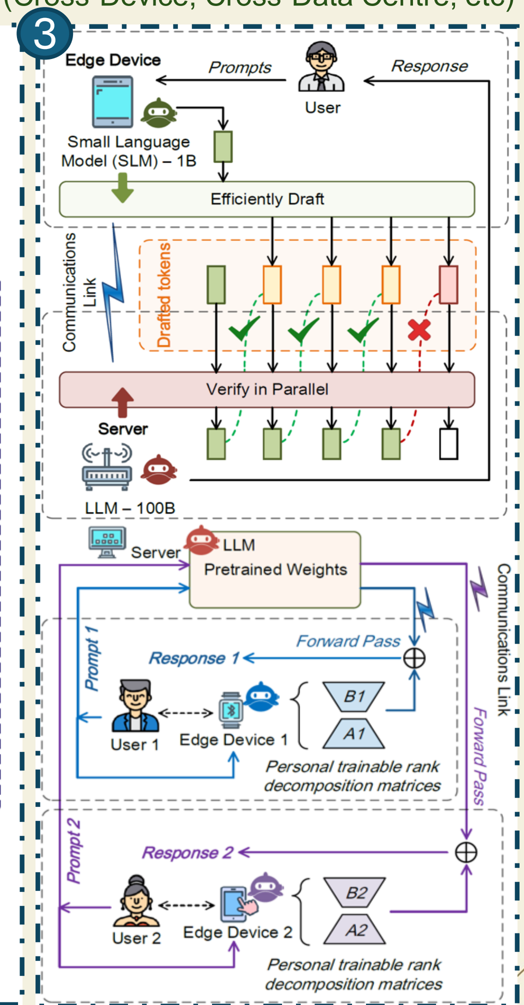
1. Distributed Diffusion Model Services



2. Edge AI-aided Personalized Network Services



3. Distributed LLM Inference (Cross-Device, Cross-Data Centre, etc)



H. Du, et al. "User-centric interactive AI for distributed diffusion model-based AI-generated content." IEEE TPAMI, major revision submitted, 2024

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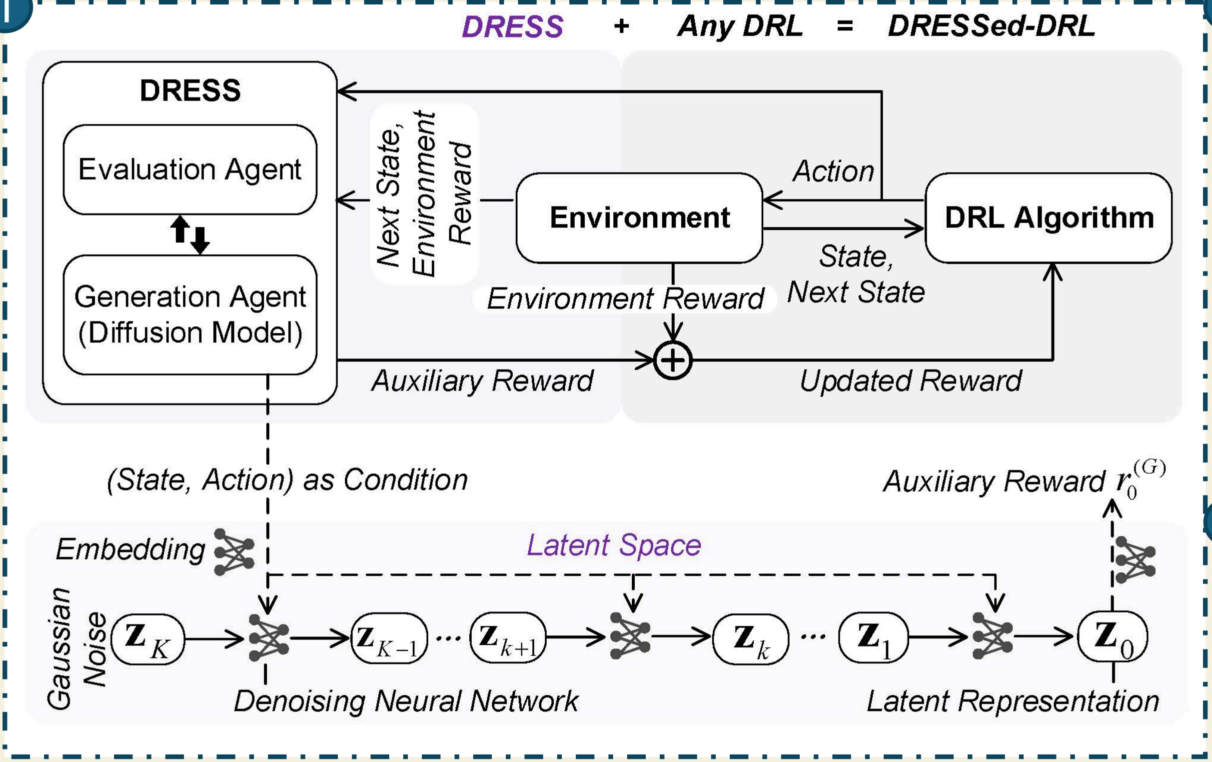
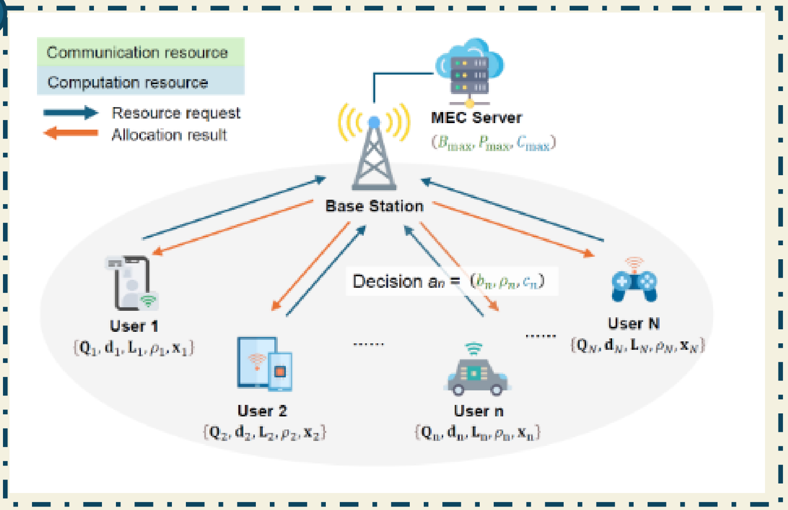
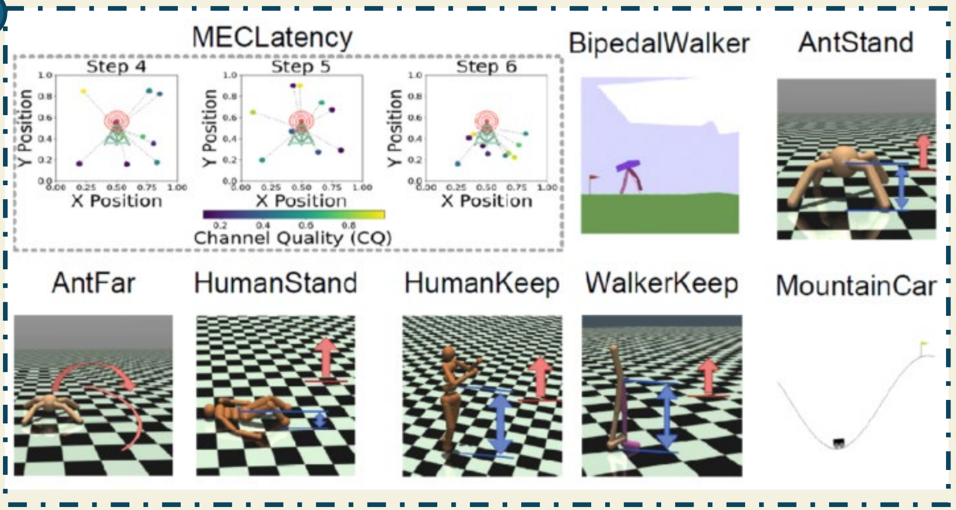
DRESS: Diffusion Reasoning-based Reward Shaping Scheme for Intelligent Networks







1. The proposed DRESS framework and its integration with various DRL architectures.
2. MECLatency: one centralized BS in management of the communication and computation resources and N users proposing resource requests.
3. Benchmark environments. MECLatency for robust wireless network optimization simulation and seven classical control tasks

You, F., Du, H., Hou, X., Ren, Y., & Huang, K. (2025). DRESS: Diffusion Reasoning-based Reward Shaping Scheme For Intelligent Networks. arXiv preprint arXiv:2503.07433.

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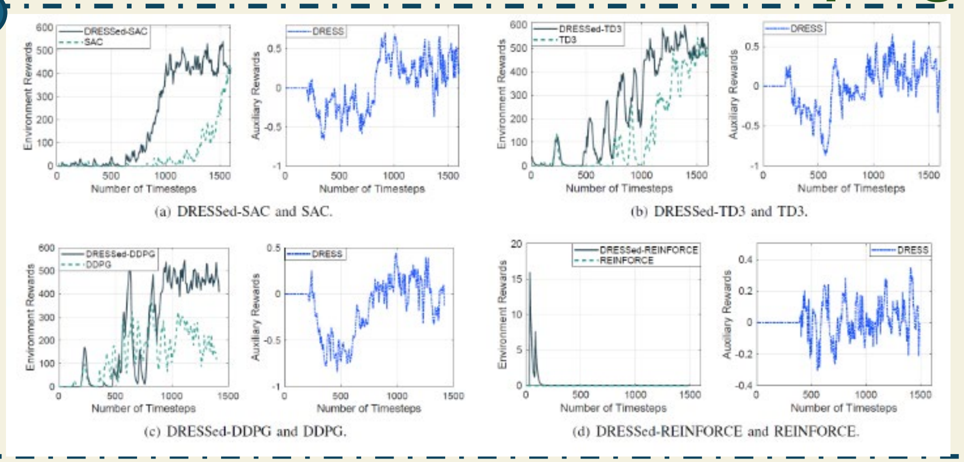
DRESS: Diffusion Reasoning-based Reward Shaping Scheme for Intelligent Networks

1

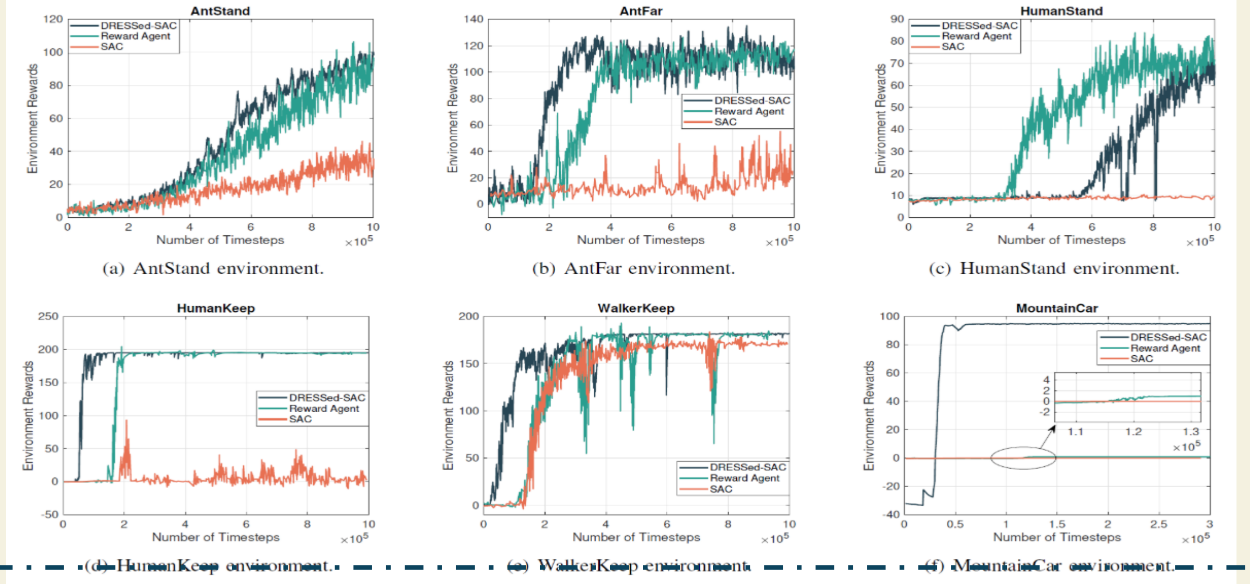


(a) DRESSed-SAC and SAC. (b) DRESSed-TD3 and TD3.
(c) DRESSed-DDPG and DDPG. (d) DRESSed-REINFORCE and REINFORCE.

1. Environment and auxiliary rewards of DRESSed-DRL and DRL algorithms in MECLatency with a latency limit of 0.02 and β = 0.2.

2. Comparison of training performance in several DRL benchmark environments across DRESSed-SAC, reward agent-based method, and SAC.

2



(a) AntStand environment. (b) AntFar environment. (c) HumanStand environment.
(d) HumanKeep environment. (e) WalkerKeep environment. (f) MountainCar environment.

Conclusion

The proposed DRESS-based DRL framework represents a practical solution to the evolving challenges of 6G wireless networks, providing a robust and efficient approach for optimizing network performance under extreme conditions.

"If your DRL's a mess, DRESS it!"

You, F., Du, H., Hou, X., Ren, Y., & Huang, K. (2025). DRESS: Diffusion Reasoning-based Reward Shaping Scheme For Intelligent Networks. arXiv preprint arXiv:2503.07433.

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Thank you!